Type MS · Apply the triplet code to bases, codons and amino acids

Answer 1 of 6 · Question 6 part c · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

6(a)	• holds DNA/chromosomes/genetic information in nucleus; • controls what enters and leaves nucleus / allows movement of substances/named substance out of/into the nucleus/exchange of substances between nucleus and cytoplasm;		**(2)**

Question Number	Answer	Additional guidance	Mark
6(b)	B CCG;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(c)	threonine;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(d)	Any six from: • tRNA has anticodons; • translation; • of mRNA; • amino acids carried by tRNA; • joined together; • by peptide bonds; • forming polypeptide/protein; • (final) protein released;	Mp3 must be in context of translation	**(6)**

Question Number	Answer	Additional guidance	Mark
6(e)	• causes change in base sequence/genetic code/named type of mutation; • of DNA;		**(2)**

Total 12 Marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 1 of 6 · Question 6 part c · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 2 of 6 · Question 6 part b · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6(a)	• holds DNA/chromosomes/genetic information in nucleus; • controls what enters and leaves nucleus / allows movement of substances/named substance out of/into the nucleus/exchange of substances between nucleus and cytoplasm;		**(2)**

Question Number	Answer	Additional guidance	Mark
6(b)	B CCG;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(c)	threonine;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(d)	Any six from: • tRNA has anticodons; • translation; • of mRNA; • amino acids carried by tRNA; • joined together; • by peptide bonds; • forming polypeptide/protein; • (final) protein released;	Mp3 must be in context of translation	**(6)**

Question Number	Answer	Additional guidance	Mark
6(e)	• causes change in base sequence/genetic code/named type of mutation; • of DNA;		**(2)**

Total 12 Marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 2 of 6 · Question 6 part b · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

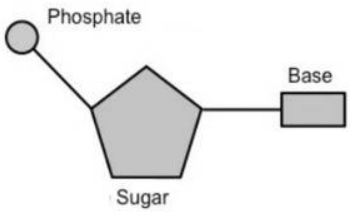
Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 3 of 6 · Question 8 part a.iv · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.iv in this mark scheme extract

8 (a) (i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1)  	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)	any three from • deletion(1)	allow description of deletion	3

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 3 of 6 · Question 8 part a.iv · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.iv in this mark scheme extract

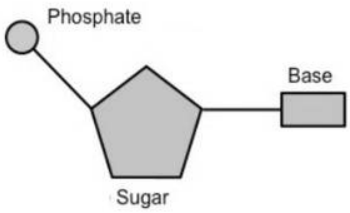
	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 4 of 6 · Question 8 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

8 (a) (i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1)  	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)	any three from • deletion(1)	allow description of deletion	3

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 4 of 6 · Question 8 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 5 of 6 · Question 1 part c · 2 marks

2023 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

Number				
1 (a) (i)	<table><tr><th rowspan="2">cell structure</th><th colspan="3">Function</th></tr><tr><th>makes proteins</th><th>contains DNA</th><th>allows substances in and out of cells</th></tr><tr><td>nucleus</td><td></td><td>✓</td><td></td></tr><tr><td>ribosomes</td><td>✓</td><td></td><td></td></tr><tr><td>cell membrane</td><td></td><td></td><td>✓</td></tr></table>	One mark for each correct row	3
(ii)	cell membrane/cell wall/cytoplasm/plasmid/DNA/ribosomes/flagellum;		1
(b) (i)	Part W phosphate (group); Part X sugar/ deoxyribose sugar;		2
(ii)	• complementary base pairing; • A-T, G-C; • hydrogen bonds between bases;		3
(c)	153 ÷ 3; 51;	Full marks for correct final answer Ecf allow correct final answer to calculation for 1 mark	2

Total for Question 1 = 11 Marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 6 of 6 · Question 5 part a · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number				
5	(a)	4^3; = 64;	Full marks for correct answer	2
	(b) (i)	• change in order/sequence; • of bases; • different protein may be formed;		3
	(ii)	uv light/λ rays/ionising radiation/X-rays;	accept any suitable example	1
	(iii)	• can't catalyse reaction/slows reaction/activity; • because can't form E-S complexes/substrate can't bind with active site/active site not complementary;		2
	(iv)	doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid;		1

Total 9 marks

Type MS · Apply the triplet code to bases, codons and amino acids

Answer 6 of 6 · Question 5 part a · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks